

#2: Using the Right Word

“Standard American English—the English of our dictionaries and grammar books—is a great, messy deluge of words, some of which overlap in meaning, many of which have multiple meanings, and many of which can be used as various parts of speech.”—Barbara Wallraff

**“To treat the sick, you must have a good knowledge of the healthy. But it is even better to know something about the disease. If the writer means to fight for the best possible use of language, he must be forever on his guard.”
—Konstantin Fedin**

(Both quoted by Bryan Garner [a major god in Ms. Particular’s pantheon] in *Garner’s Modern American Usage* [Oxford University Press, 2003])

The purpose of this exercise is to give you practice in recognizing words that are frequently misused, misspelled, or confused for one another in physics papers. All the examples are taken from recent papers published in *Physical Review Letters*. (Honestly; I’m not making these up.) Bibliographic citations have been omitted to protect the guilty.

In this exercise, use the “Review” and “Track Changes” tools in Word to replace the wrong words with correct ones. Here’s an example of what you should do:

The repeated application of hydrogen peroxide ~~lead~~led to significant corrosion of the junctions.

You may wish to review some of the *Ms. Particular* micro-lectures on the course website for guidance.*

1. Our results compliment and extend previous work based on conformal field theory.

A: Our results **complement** and extend previous work based on conformal field theory.

2. The ion fluid exhibits an alternance from azimuthal to radial velocity due to the electric field inside the rotating structure. [Please replace the unfortunate “due to,” which Ms. Particular hates with a white-hot passion; q.v. <http://people.physics.illinois.edu/Celia/MsP/DueTo.pdf>.]

A: The ion fluid exhibits an **alternate** from azimuthal to radial velocity due to the electric field inside the rotating structure.

* Hint: Review “accord,” “alternate,” “complement,” “discreet,” “dramatic,” “inherent,” and “principal.” [And note that in US usage, commas and periods go *inside* [quotation marks](#).]

3. We describe a proof-of-principal experiment demonstrating the use of spread spectrum technology at the single photon level.

A: We describe a proof-of-**principle** experiment demonstrating the use of spread spectrum technology at the single photon level.

4. However, for the sake of this discussion, we assume that they are combined into discreet energy bins such that each j_ω is of the order of j .

A: However, for the sake of this discussion, we assume that they are combined into **discrete** energy bins such that each j_ω is of the order of j .

5. Such a profile can also be formed by the separation of the cold and hot electron populations as described by two temperature plasma expansion theory and these processes may complement one another. [Look for the missing hyphens and insert the absolutely mandatory comma.]

A: Such a profile can also be formed by the separation of the cold and hot electron populations -- as described by two temperature plasma expansion theory -- and these two processes may **complement** one another.

6. We discuss the physical realizability of our results for thin molecular and complex fluid films, and predict that such phenomenon can in principal be observed in various flow geometries.

A: We discuss the physical realizability of our results for thin molecular and complex fluid films and predict that such phenomenon can, **in principle**, be observed in various flow geometries.

7. Surprisingly, the turbulent GAM dispersion relation is qualitatively equivalent to the linear GAM dispersion but can have drastically enhanced group velocities. [This sentence is doubly bad because the acronym, used *twice*, is not defined—and the sentence is from the abstract!]

A: Surprisingly, the turbulent GAM dispersion relation is equivalent to the linear GAM dispersion, but can have dramatically enhanced group velocities

8. The number of Dirac eigenvalues below some bound is an integer, and jumps at discreet points in torus moduli space. [Why is the comma incorrect?]

A: The number of Dirac eigenvalues below some bound is an integer (**removed comma**) and jumps at discreet points in torus moduli space.

9. We have used an alternate technique based on electron diffraction to determine $|b|$.

A: We have used an **alternative** technique -- **based on electron diffraction** -- to determine $|b|$.

10. Although both relaxation regimes are innately nonlinear, these regimes are unified and their positions along the frequency axis are set by the imposed strain-rate amplitude. [Insert the missing comma.]

A: Although both relaxation regimes are innately nonlinear, these regimes are unified, and their positions along the frequency axis are set by the imposed strain-rate amplitude.

11. In this system, in accordance to Kohn's theorem, strong reduction of the energy level separation caused by the electron-electron interaction compensates the depolarization shift.

A: In this system, in agreement with Kohn's theorem, a strong reduction of the energy level separation -- **caused by the electron-electron interaction** -- compensates for the depolarization shift.

12. In order for both AFD and AFE orders to coexist, the competition between them is reconciled through the formation of the super structure where both instabilities are alternatively interwoven.

A: In order for both AFD and AFE orders to coexist, the competition between them is reconciled through the formation of the super structure -- **where both instabilities are alternatively interwoven**.

Upload your exercise to gradescope at the end of today's workshop.